Question:

Four days after undergoing an elective total hip replacement, a 65-year-old woman develops a DVT that embolizes to the lung. Along with tachypnea, tachycardia, and cough, the patient would most likely present with a PaO2 of what?Answer Choices:A. 120 mmHgB. 110 mmHgC. 85 mmHgD. 60 mmHg

Reasoning + Answer:

<think>### Finding Reasoning Paths: 1. **Total hip replacement âĨĴ Blood clots âĨĴ Chronic venous insufficiency âĨĴ Lower extremity venous thrombosis (DVT) âĨĴ Pulmonary embolism âĨĴ Caused by turbulent airflow âĨĴ Obstructive hypoxemia âĨĴ PaO2 reduction. **2. ** Cough + unilateral leg swelling + femoral vein x-ray abnormality âĨĴ Lower extremity DVT âĨĴ Pulmonary embolism âĨĴ PaO2 reduction. **3. ** Lower extremity DVT âĨĴ Acute embolism âĨĴ Impaired alveolar ventilation âĨĴ Obstructive hypoxemia âĨĴ PaO2 reduction. **4. ** DVT âĨĴ PFT finding of reduced residual volume âĨĴ Right atrium involvement âĨĴ Pulmonary embolism âĨĴ PaO2 reduction. **---

Reasoning Process: 1. **Understanding the clinical scenario: ** - A 65-year-old woman presents four days after total hip replacement with tachypnea, tachycardia, and cough. These symptoms suggest significant respiratory distress. - A DVT embolizes to the lung, which indicates a pulmonary embolism (PE). Common precipitants for PE include recent surgery, immobility, trauma, recent travel, malignancy, or immobilization. 2. **Common precipitants for pulmonary embolism: ** - Common causes of PTE include ** dyslipidemia**, ** cancer**, ** hyperhomocysteinemia**, ** hyperviscosity syndromes**, and ** family history of venous thromboembolism (VTE)**. - Dyslipidemia is strongly associated with increased risk of both VTE and PE, making it a likely culprit in this case. 3. **Pathophysiology of pulmonary embolism: ** - A pulmonary embolism typically manifests with symptoms such as pleuritic chest pain, dyspnea, hemoptysis, and tachypnea. The physical findings of ** unilateral leg swelling** and ** femoral vein abnormality** on x-ray confirm the diagnosis of DVT. - PE leads to impaired alveolar ventilation in the involved lung segment due to obstruction of blood flow. This results in reduced oxygen extraction and alveolar dead space, causing hypoxemia (reduced PaO2) and increased ventilation-perfusion mismatch. 4. **Physiologic findings and physiology: ** - A PFT finding of ** reduced residual volume** suggests right atrium involvement due to blood flow obstruction in the affected lung segment. This confirms the diagnosis of a large embolism causing PE. - Additionally, the typical features of PFT (reduced VC and CO) reflect the profound hypoxemia and ventilatory compromise associated with PTE. 5. **Relevance of other differential diagnoses: ** - Elevated LFT or CK-MB typically indicate myocardial infarction but are not relevant here as this patient presents with pulmonary findings, not cardiovascular symptoms. - A diagnosis of gallbladder perforation is not supported by the clinical presentation of a DVT embolizing to the lung. ---

Conclusion: The patient's dyspnea, tachypnea, and tachycardia suggest ** obstructive hypoxemia**, likely due to a ** Pulmonary Embolism** (large embolus involving the lung). A PFT finding of ** reduced residual volume** further confirms this diagnosis. PFT findings include ** reduced residual volume** due to right atrium involvement, ** reduced VC** due to ventilation-perfusion mismatch, and ** reduced CO** due to impaired alveolar ventilation. The associated symptoms and imaging findings (PE embolization causing DVT) support this conclusion. </think>

<answer>60 mmHg </answer>